

Dr. Kendall Sullivan

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🌐 SciX library with all first- and co-authored publications

Employment

- Jan. 2026 – present 📌 **Research Fellow in Exoplanets**, Mullard Space Science Laboratory, Department of Space and Climate Physics, University College London
Supervisor: Prof. Vincent Van Eylen
- Sept. 2023 – Dec. 2025 📌 **Postdoctoral Scholar**, Department of Astronomy & Astrophysics, University of California Santa Cruz
Supervisor: Prof. Natalie Batalha
- Sept. 2020 – Aug. 2023 📌 **NSF Graduate Research Fellow**, Department of Astronomy, University of Texas at Austin
PhD Advisor: Prof. Adam Kraus
- Jan. 2018 – Aug. 2018 📌 **Post-Baccalaureate Researcher**, Lowell Observatory
Supervisor: Dr. Lisa Prato

Education

- Aug. 2023 📌 **Ph.D., Astronomy**
University of Texas at Austin
Revealing Star and Planet Formation With Stellar Multiplicity
- Feb. 2018 📌 **B.S., Physics/Astronomy, cum laude**
University of Massachusetts Amherst
Commonwealth Honors College Scholar with Great Distinction
Departmental Honors in Astronomy
S and VV Coronae Australis: Two Extreme Young Binary Systems

Awards and Fellowships

- 2023 📌 **Rodger Doxsey Dissertation Award**, American Astronomical Society
- 2020 📌 **Board of Visitors Second Year Defense Award**, University of Texas at Austin
📌 **Graduate Research Fellowship (5 years)**, National Science Foundation
- 2018 📌 **Dean's Unrestricted Fellowship (5 years)**, University of Texas at Austin
📌 **Astronomy Dept. Recruitment Fellowship (2 summers)**, University of Texas at Austin

Select Invited and Contributed Talks

- May 2026 📌 Queen Mary University of London (invited seminar)
- Apr. 2026 📌 University of Surrey (invited colloquium)
- Feb. 2026 📌 UCL/Mullard Space Science Laboratory (invited colloquium)
- Jan. 2026 📌 London Exoplanet Meeting (contributed talk)
- Nov. 2025 📌 Bay Area Exoplanet Meeting (contributed talk)
- Oct. 2025 📌 University of California Santa Cruz (seminar)
- Sept. 2025 📌 University of Texas at Austin (seminar)
- Oct. 2024 📌 University of Wisconsin – Madison (invited seminar)

Select Invited and Contributed Talks (continued)

Sept. 2024	■ Yale University (invited seminar)
	■ Keck Science Meeting (contributed plenary)
June 2024	■ Cool Stars 22 (contributed plenary)
April 2024	■ Penn State/Center for Exoplanets and Habitable Worlds (invited seminar)
March 2024	■ Extreme Solar Systems V (contributed plenary)
Feb. 2023	■ Jet Propulsion Lab Exoplanet Seminar (invited seminar)
Jan. 2023	■ Dissertation Talk, AAS 241 (contributed)
	■ NASA ExoPAG 27 Early Career Symposium (contributed plenary)
Oct. 2022	■ University of Hawai'i/IfA (invited seminar)
2018-2022	■ 4 UT Austin Stars, Planets, and ISM Seminar talks

PI Observing Programs

NOIRLab/NN-Explore	■ WIYN 3.5m/NEID - 7h, 2025A
Lick Observatory	■ Shane 3-m/ShaneAO + ShARCS - 18 nights, 2024A+B, 2026A
	■ Automated Planet Finder - 100 hours, 2024B
	■ Nickel 1-m/CCD-C2 - 20 nights, 2024A
McDonald Observatory	■ Hobby-Eberly Telescope/LRS2 - 275 hours, 6 trimesters over 2021-2023 (PI; 163 h); 2025-2,-3, 2026-1,-2 (science PI; 112 h)
NOIRLab/NOAO	■ Gemini South/GMOS - 2.2 hours, 2021B
	■ IRTF/iSHELL - 4 half nights, 2019B
	■ KPNO/WIYN 0.9m telescope - 2 nights, 2017/2018

Service

AAS Journals; A&A; Nature	■ Referee, 2023 – present
UCL	■ Habitable Worlds Observatory UK Community Workshop SOC and LOC, June 2026
NASA	■ Grant Review Panel Executive Secretary, 2 years
UC Observatories, NOIRlab	■ Time Allocation Committee, multiple semesters
UT Austin	■ Graduate-Undergraduate Mentoring Program co-lead, 2021–2023
	■ TAURUS/REU weekly seminar lead organizer, 2 years

Teaching Experience

UC Santa Cruz	■ ASTR 112 (Physics of Stars) instructor, Spring 2025
	■ ASTR 19 (Introduction to Python) guest lecturer, Feb. 2024
UT Austin	■ Graduate Student Observing Seminar lecturer, Feb. 2023
	■ TAURUS/REU Introduction to Astronomy Jargon Seminar developer and lead, 2 summers

Mentored Students

UCL	■ Ross Dobson, 4th year PhD student (interim advisor), March 2026 – July 2026
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Mentored Students (continued)

UC Santa Cruz	■	Emery Grahill-Bland , 1st year PhD student (interim advisor), March 2026 – July 2026
	■	Marina Ventikos , 2nd year PhD student (interim advisor), March 2026 – July 2026
	■	Francesca Waines , 3rd year PhD student (interim advisor), March 2026 – July 2026
	■	3 MSc students (interim advisor), March 2026 – July 2026
	■	Bridget Gale , undergraduate research, Oct. 2025 – present
	■	Astrophel Fong , Lamat scholar, June 2025 – Aug. 2025
	■	Nina Carrillo , undergraduate research, senior thesis, Oct. 2024 – June 2026
	■	Ira Gupta , undergraduate research, Oct. 2024 – June 2025
	■	2 ASTR 9 groups , 4-6 students, Winter + Spring quarters 2024 + 2025
	■	Max Clark , Lamat scholar, University of Wisconsin - Madison undergraduate, June 2024 – Dec. 2024
UT Austin	■	Nathanael Burns-Watson , UT REU, Case Western Reserve University senior thesis, UT PhD student (co-advisor), June 2023 – present
	■	Hunter Brooks , Northern Arizona University undergraduate, Jan. 2022 – Feb. 2023

Outreach

Nature	■	News and Views author: <i>Dying stars give a second wind to exoplanet formation</i> , May 2025
UC Santa Cruz	■	New Discoveries About Planets Outside our Solar System , Public Talk, Orwell Free Library, Orwell VT, Dec. 2025
	■	Astronomy on Tap speaker , May 2025
	■	Ask an Astronomer , 2024–2025

Collaborations

- **Transits in the Roman Galactic EXoplanet Survey (TRExS)**, 2025 – present
- **NASA ExEP SIG2 (Exoplanet Demographics)**, 2023–present

Publications (First-author = 12; * shows a student publication.)

- 1 N. Burns-Watson*, K. **Sullivan**, and A. L. Kraus, “Determining the Host Stars of Planets in Binary Star Systems with Asterodensity Profiling: Investigating the Canonical Radius Gap”, *AJ* **171**, 327, 327 (2026).
- 2 J. Levine, C. Manea, K. Hawkins, K. **Sullivan**, K. H. R. Rubin, Z. Maas, and A. C. Nine, “Barium Stars across the Milky Way: Probing Their Origins via the GALAH Survey”, *ApJ* **1000**, 117, 117 (2026).
- 3 E. Newton, M. Rice, K. **Sullivan**, J. Christiansen, S. Blunt, and S. Sagynbayeva, “The Role of Stellar Multiplicity in the Prevalence of Small, Cool Planets”, in *ASTRONOMICAL SOCIETY OF THE PACIFIC CONFERENCE SERIES*, Vol. 542, edited by J. C. Lee, J. Noviello, S. LaMassa, and M. Postman, Astronomical Society of the Pacific Conference Series (Feb. 2026), p. 457.
- 4 K. **Sullivan**, A. Dattilo, and N. M. Batalha, “The First Direct Occurrence Rate Estimates for Kepler Exoplanets in Small-separation Binary Star Systems: Planet Occurrence Is Suppressed in Binary Stars”, *AJ* **171**, 53, 53 (2026).
- 5 K. **Sullivan** and G. J. Gilbert, “Planetary Architectures of Kepler Compact Multiplanet Systems with Binary-star Companions”, *AJ* **171**, 299, 299 (2026).
- 6 R. B. Fernandes, S. Johnson, G. J. Bergsten, S. Bhure, K. M. Boley, A. P. Boss, S. Bryson, W. DeRocco, J. Dietrich, A. Duck, S. Giacalone, A. F. Gupta, M. Y. He, M. Kunitomo, K. Ment, S. Sagar, M. L. Silverstein, K. **Sullivan**¹, E. H. Vrijmoet, K. Wagner, R. F. Wilson, L. Brefka, R. Belikov, A. Chakrabarty, J. L. Christiansen, D. R. Ciardi, A. Dattilo, E. Fitzmaurice, E. B. Ford, A. Hotnisky, S. Jones, A. Kar, R. Kopparapu, N. Lowson, E. E. Mamajek, B. Mennesson, M. R. Meyer, S. Millholland, G. D. Mulders, S. E. Mullally, A. Murlidhar, I. Pascucci, D. Ragozzine, P. Robertson, K. Stapelfeldt, and J. Wright, “Are We There Yet? Challenges in Quantifying the Frequency of Earth Analogs in the Habitable Zone”, *PASP* **137**, 121001, 121001 (2025).
- 7 K. **Sullivan**, A. L. Kraus, T. A. Berger, and D. Huber, “Quantifying the Contamination from nearby Stellar Companions in Gaia DR3 Photometry”, *AJ* **169**, 29, 29 (2025).
- 8 M. Clark*, K. **Sullivan**, and M. Soares-Furtado, “Moderate-separation Binary Companions May Influence Young Stellar X-Ray Luminosity”, *Research Notes of the American Astronomical Society* **8**, 318, 318 (2024).
- 9 E. L. Evans, T. J. Dupuy, K. **Sullivan**, A. L. Kraus, D. Huber, M. J. Ireland, M. Ansdell, R. L. Kuruwita, R. A. Martinez, and M. L. Wood, “Orbital architectures of planet-hosting binaries - III. Testing mutual inclinations of stellar and planetary orbits in triple-star systems”, *MNRAS* **534**, 575–607 (2024).
- 10 E. Gaidos, C. A. Clark, and K. **Sullivan**, “The Mass–Radius Relation for Planets in Binary Systems”, *Research Notes of the American Astronomical Society* **8**, 319, 319 (2024).
- 11 K. **Sullivan** and A. L. Kraus, “Starspots and Undetected Binary Stars Have Distinct Signatures in Young Stellar Associations”, *ApJ* **969**, 117, 117 (2024).
- 12 K. **Sullivan**, A. L. Kraus, T. A. Berger, T. J. Dupuy, E. Evans, E. Gaidos, D. Huber, M. J. Ireland, A. W. Mann, E. A. Petigura, P. C. Thao, M. L. Wood, and J. Zhang, “Revising Properties of Planet–Host Binary Systems. IV. The Radius Distribution of Small Planets in Binary Star Systems Is Dependent on Stellar Separation”, *AJ* **168**, 129, 129 (2024).

¹Co-first-author

- 13 K. **Sullivan**, A. L. Kraus, D. Huber, E. A. Petigura, E. Evans, T. Dupuy, J. Zhang, T. A. Berger, E. Gaidos, and A. W. Mann, “Revising Properties of Planet-Host Binary Systems. III. There Is No Observed Radius Gap for Kepler Planets in Binary Star Systems”, *AJ* **165**, 177, 177 (2023).
- 14 K. **Sullivan** and A. L. Kraus, “Optical and Near-infrared Excesses are Correlated in T Tauri Stars”, *ApJ* **928**, 134, 134 (2022).
- 15 K. **Sullivan** and A. L. Kraus, “Revising Properties of Planet-Host Binary Systems. II. Apparent Near-Earth-analog Planets in Binaries Are Often Sub-Neptunes”, *AJ* **164**, 138, 138 (2022).
- 16 K. **Sullivan**, A. L. Kraus, and A. W. Mann, “Revising Properties of Planet-Host Binary Systems. I. Methods and Pilot Study”, *ApJ* **935**, 141, 141 (2022).
- 17 L. A. Nofi, C. M. Johns-Krull, R. López-Valdivia, L. Biddle, A. S. Carvalho, D. Huber, D. Jaffe, J. Llama, G. Mace, L. Prato, B. Skiff, K. R. Sokal, K. **Sullivan**, and J. Tayar, “Projected Rotational Velocities and Fundamental Properties of Low-mass Pre-main-sequence Stars in the Taurus-Auriga Star-forming Region”, *ApJ* **911**, 138, 138 (2021).
- 18 K. **Sullivan** and A. L. Kraus, “Undetected Binary Stars Cause an Observed Mass-dependent Age Gradient in Upper Scorpius”, *ApJ* **912**, 137, 137 (2021).
- 19 L. Flagg, C. M. Johns-Krull, L. Nofi, J. Llama, L. Prato, K. **Sullivan**, D. T. Jaffe, and G. Mace, “CO Detected in CI Tau b: Hot Start Implied by Planet Mass and M_K ”, *ApJL* **878**, L37, L37 (2019).
- 20 K. **Sullivan**, L. Prato, S. Edwards, I. Avilez, and G. H. Schaefer, “S and VV Corona Australis: Spectroscopic Variability in Two Young Binary Star Systems”, *ApJ* **884**, 28, 28 (2019).
- 21 J. A. Irwin, P. Schmidt, A. Damas-Segovia, R. Beck, J. English, G. Heald, R. N. Henriksen, M. Krause, J.-T. Li, R. J. Rand, Q. D. Wang, T. Wiegert, P. Kamienieski, D. Paré, and K. **Sullivan**, “CHANG-ES - VIII. Uncovering hidden AGN activity in radio polarization”, *MNRAS* **464**, 1333–1346 (2017).